

OS Unit 1 MSE

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Question: Introduction

Answer:

Operating System ek software hota hai jo computer ke hardware aur user ke beech bridge ki tarah kaam karta hai. Ye applications ko run karne, memory handle karne, files manage karne aur input/output devices (jaise printer, disk drive) ko control karne me help karta hai. Har computer ko ek OS zaroor chahiye hota hai, tabhi browsers, MS Office, Notepad, games jaise apps properly chal paate hain. Is blog me hum OS ka development aur uski journey simple tarike se samjhenge.

Question: What do you mean by Operating System?

Answer:

Operating System ek set of programs hota hai jo computer ke hardware ko manage karta hai aur user ko system use karne ka easy way deta hai. Har computer, chahe chhota ho ya bada, OS ke bina kaam nahi kar sakta. User directly hardware se baat nahi karta. Agar user ko hardware se koi kaam karwana ho, to wo OS ke through request karta hai. OS hi user aur hardware ke beech connection banata hai.

Question: What is Evolution of Operating Systems?

Answer:

Operating Systems time ke saath kaafi upgrade hue hain. Pehle systems slow aur expensive hote the, aaj ke OS fast, powerful aur affordable hain. OS evolution ko 4 generations me divide kiya jata hai:

- First Generation
- Second Generation
- Third Generation
- Fourth Generation

First Generation (1945–1955): Serial Processing

Is time computers me operating system hota hi nahi tha. Programmer ko machine ko directly instructions dene padte the. Ye electronic computers ki shuruaat thi, jo mechanical computers se zyada fast the.

Is generation me sabse bada issue scheduling aur setup time ka tha. Compiler load karna, program save karna, link karna sab manually hota tha. Agar beech me error aaya to poora process dobara start karna padta tha.

Example: Windows 95 aur 98 ko serial processing concept ka simple example mana jata hai.

Question: Second Generation (1955–1965) – Batch System

Answer:

Second generation me **batch processing system** use hota tha. Isme jobs ko cards aur tapes par store karke ek sequence me run kiya jata tha. Yaha complete operating system nahi tha, par FMS aur IBSYS jaise OS features available the.

Program punch cards par likhe jate the, tape me transfer hote the, aur computer ek job khatam karte hi next job automatically start karta tha. Ye system user ke liye thoda inconvenient tha, par computer ko maximum busy rakhne ke liye use hota tha.

Memory protection aur timer jaisa safety system bhi use hota tha. Lekin jab input/output devices busy hote the, tab CPU idle rehta tha, isliye CPU utilization weak hota tha.

Example: IBM ka MVS OS batch processing ka example hai.

Question: Fourth Generation (1980–Now)

Answer:

Fourth generation me computers networked ho gaye aur users ko **GUI** mila, jisse computer use karna easy ho gaya. Is time **time-sharing OS** aur **Macintosh OS** popular hue.

1) Time-Sharing Operating System

Is OS me multiple users ek hi system ko same time par use kar sakte hain. Processor apna time small parts me divide karke sab users ko deta hai. Pehle command-line terminals use hote the, jaha user text commands likh kar interact karta tha. Fast server ek hi waqt me bohot saare users ke tasks handle kar leta tha.

Example: Unix OS.

2) Macintosh Operating System

Mac OS graphical systems par ki gayi research ka result tha. Pehle GUI hardware bahut expensive tha, par technology improve hone ke baad Macintosh commercially successful ho gaya. Sketchpad jaise early GUI concepts iske base the.

Example: Mac OS X 10.6.8.

Question: Batch Operating System

Answer:

Batch OS me user directly computer se interact nahi karta. Ek operator similar type ke jobs ko group karke batch banata hai aur system unhe sequence me execute karta hai.

Advantages:

- Processor ko approx idea hota hai ki job kitna time legi.

- Multiple users batch system use kar sakte hain.
- Idle time bahut kam hota hai.
- Repeated large work manage karna easy hota hai.

Disadvantages:

- Operator ko batch system ka good knowledge chahiye.
- Debugging mushkil hoti hai.
- Kabhi-kabhi system costly hota hai.
- Agar ek job fail ho jaye, to baaki jobs ko wait karna padta hai.

Example: Payroll systems, bank statements.

Question: Multi-Programming Operating System**Answer:**

Multiprogramming OS me main memory me ek se zyada programs rakhe jaate hain, aur CPU unme se kisi bhi program ko run kar sakta hai. Ye system resources ka better use karne ke liye banaya gaya hai.

Advantages:

- System ka throughput badhta hai.
- Response time kam hota hai.

Disadvantages:

- User directly system resources ke sath interact nahi kar sakta.

Question: Multi-Processing Operating System**Answer:**

Multi-processing OS me ek system me ek se zyada CPU hote hain, jo saath me kaam karke performance aur throughput ko improve karte hain.

Advantages:

- System ka throughput kaafi badh jata hai.
- Agar ek CPU fail ho jaye, to doosra processor kaam continue kar sakta hai.

Disadvantages:

- Multiple CPUs hone ki wajah se system thoda complex aur samajhne me mushkil ho sakta hai.

Question: Multi-Tasking Operating System**Answer:**

Multitasking OS basically multiprogramming system hota hai jo **Round-Robin scheduling** use karta hai. Ye

ek hi time multiple programs ko run karne deta hai.

Types:

- Preemptive Multitasking
- Cooperative Multitasking

Advantages:

- Multiple programs ek saath run ho sakte hain.
- Memory management better hota hai.

Disadvantages:

- Heavy programs zyada chalne par system garam ho sakta hai.

Question: Time-Sharing Operating System

Answer:

Time-sharing OS me har task ko thoda-thoda CPU time diya jata hai, jisse sab tasks smoothly run hote hain. Multiple users ek hi system ko ek saath use kar sakte hain. Har task ko milne wala time **quantum** kehlata hai. Quantum khatam hote hi OS next task par switch ho jata hai.

Advantages:

- Har task ko equal chance milta hai.
- Software duplication ke chances kam hote hain.
- CPU idle time reduce hota hai.
- Resources (CPU, memory, devices) share hote hain, cost kam padti hai.
- Users ek saath kaam kar sakte hain, productivity badhti hai.
- Interactive environment milta hai, user experience better hota hai.

Disadvantages:

- Reliability issues aa sakte hain.
- Security aur data integrity ka dhyan rakhna padta hai.
- Data communication problems ho sakti hain.
- Scheduling aur switching ke karan overhead zyada hota hai.
- System complex hota hai, bugs ka risk badh jata hai.
- Multiple users hone se security risks bhi badhte hain.

Examples:

- **IBM VM/CMS:** 1972 me launch hua, virtual machine environment deta hai jaha multiple users apna OS run kar sakte hain.
- **TSO (Time Sharing Option):** IBM System/360 ke liye banaya gaya tha, jisme users ek hi computer ko ek saath use kar sakte the.

- **Windows Terminal Services:** Users remotely Windows server ko access karke apne apps real-time me run kar sakte hain.

Question: Distributed Operating System

Answer:

Distributed OS me kai independent computers ek shared network ke through connected hote hain. Har system ka apna CPU aur memory hota hai, isliye inhe **loosely coupled systems** bhi bola jata hai. Users network me kisi dusre computer par stored files ya software ko remotely access kar sakte hain.

Advantages:

- Ek system fail ho jaye to bhi network par koi farak nahi padta.
- Email jaisi services se data exchange fast hota hai.
- Shared resources ki wajah se computation fast aur reliable hota hai.
- Host computer ka load kam hota hai.
- Network me naya system add karna easy hota hai.
- Data processing me delay kam hota hai.

Disadvantages:

- Agar main network fail ho jaye to pura communication ruk jata hai.
- Distributed systems ke liye programming languages abhi fully defined nahi hain.
- Systems aur software dono costly aur complex hote hain.

Example: LOCUS.

Question: Network Operating System

Answer:

Network OS server par run hota hai aur data, users, security, applications aur network resources ko manage karta hai. Ye systems files, printers aur applications ko ek chhote private network me share karne ki facility dete hain. Users network ki configuration aur connections ke baare me aware hote hain, isliye ise **tightly coupled system** bhi kaha jata hai.

Advantages:

- Centralized server system stable hota hai.
- Security server ke through manage hoti hai.
- New hardware aur technologies ko add karna easy hota hai.
- Remote access possible hoti hai.

Disadvantages:

- Servers kaafi costly hote hain.

- Zyada operations ke liye central server par depend rehna padta hai.
- Regular maintenance aur updates ki zarurat hoti hai.

Examples: Windows Server 2003/2008, UNIX, Linux, Mac OS X, Novell NetWare, BSD.

Question: Real-Time Operating System

Answer:

Real-Time OS un systems me use hota hai jaha input ka response time bahut kam chahiye hota hai. Inka use missile control, air-traffic systems, robots jaise critical areas me hota hai.

Types:

- **Hard Real-Time OS:** Time limit bahut strict hoti hai. Chhoti si delay bhi risky hoti hai, jaise airbags ya automatic parachute systems. Inme virtual memory use nahi hoti.
- **Soft Real-Time OS:** Time requirement thodi flexible hoti hai. Thoda delay tolerate ho sakta hai.

Advantages:

- Devices ka maximum use hota hai, output zyada milta hai.
- Task switching bahut fast hoti hai (microseconds me).
- Running application par full focus hota hai.
- Embedded systems me bhi easily use ho sakta hai.
- Systems almost error-free hote hain.
- Memory allocation acchi tarah manage hoti hai.

Disadvantages:

- Ek time par limited tasks hi run ho sakte hain.
- High-performance hardware chahiye hota hai, jo costly hotta hai.
- Algorithms complex hote hain.
- Fast device drivers aur interrupt signals ki zarurat hoti hai.
- Thread priority handling weak hoti hai.

Examples: Scientific experiments, medical imaging, industrial control, weapons, robots, air-traffic control.

Question: Views of Operating System

Answer:

Operating System ek framework hota hai jo user ke applications ko hardware se connect karta hai. OS khud koi direct kaam nahi karta, but ek environment deta hai jisme apps smoothly kaam kar sakti hain. OS ko do tarah se dekha ja sakta hai: **User View** aur **System View**.

1. User View

User view is baat par based hota hai ki user system se kaise interact karta hai.

Kuch systems single user ke liye bante hain jaha ease of use important hoti hai. Kuch me multiple users ka load hota hai, jaha performance aur resource use important ho jata hai.

User View ke types:

- **Single User Viewpoint**

Yaha system ek user ke liye bana hota hai. Focus accessibility par hota hai, resource sharing par nahi.

Example: normal PC, laptop.

- **Multiple User Viewpoint**

Yaha ek main computer (mainframe/server) ko kai users access karte hain. CPU ko memory aur resources efficiently divide karne padte hain. Example: client-server systems.

- **Handheld User Viewpoint**

Smartphones aur tablets jaha touch interface hota hai. Ye fully powerful nahi hote PCs ki tarah, but user view par heavy focus hota hai.

- **Embedded System User Viewpoint**

Embedded systems me user interaction bahut limited hota hai. Example: TV remote, microwave panel. Yaha user sirf simple commands deta hai.

In short:

User view batata hai ki user OS ko kaise feel karta hai, aur system view batata hai ki hardware OS ke through tasks kaise handle karta hai.

Question: System View

Answer:

System view me OS ko ek **resource allocator** ke roop me dekha jata hai. Computer ke paas hardware aur software resources hote hain, jinko OS efficiently manage karta hai. OS ka main goal yaha performance ko maximize karna hota hai.

System view me user interaction se zyada focus is baat par hota hai ki hardware aur OS kaise milkar tasks complete karte hain.

1. Resource Allocation

Hardware me registers, RAM, ROM, cache, CPU, I/O devices jaise resources hote hain. Jab programs inki demand karte hain, to OS decide karta hai ki konsa resource kisko milega.

OS paging, virtual memory, caching jaisi techniques use karke performance improve karta hai. Agar resource allocation weak ho jaye to system slow ya hang ho sakta hai.

2. Control Program

Control program I/O devices aur OS ke beech communication manage karta hai. Jab user koi action maangta

hai jo input-output device se related ho, to OS device ko control karta hai, detect karta hai aur handle karta hai.

System view basically batata hai ki hardware OS ke control me kaise efficiently kaam karta hai.

Question: Process

Answer:

Process ek **program in execution** hota hai. Matlab jo program file me likha hota hai, wo run hote hi process ban jata hai aur sequential steps me execute hota hai. Process system ka basic unit of work hota hai.

Jab program memory me load hota hai, to uska structure 4 parts me divide hota hai:

- **Stack** – function calls aur local variables
- **Heap** – dynamic memory
- **Text** – program code
- **Data** – global variables

Ye layout batata hai ki process memory me kaise organize hota hai.

Question: Program Execution

Answer:

OS decide karta hai ki program kaise execute hoga. Pehle program memory me load hota hai, phir CPU scheduling algorithms (jaise FCFS, SJF) ke according run hota hai. Execution ke time OS deadlock avoid karta hai aur ensure karta hai ki user aur system programs smoothly chal sake. Efficient execution ke liye OS available resources ka best use karta hai.

Input/Output Operations

OS input-output operations handle karta hai. Device drivers ke through devices aur user ke beech communication establish hota hai. Jab kisi program ko I/O device chahiye hota hai, OS usko access provide karta hai.

Communication Between Processes

Different processes ke beech data transfer OS manage karta hai. Agar processes alag computers par ho aur network ke through connected ho, tab bhi unki communication ka control OS hi sambhalta hai.

Question: File Management

Answer:

OS files ko manage karta hai. Jab kisi program ko file chahiye hoti hai, OS usko permission deta hai, jaise

read-only, read-write, etc. User ko files create, delete aur organize karne ka option bhi OS deta hai. OS decide karta hai ki data kaha store hoga (hard disk, pen drive, etc.) aur kaise store hoga.

Question: Memory Management

Answer:

Memory management ko cricket example se samajh sakte hain. Jaise team manager decide karta hai kaun playing 11 me hoga, waise hi OS check karta hai ki new program ko memory milegi ya nahi. Agar program eligible ho, to OS decide karta hai kitni memory lagegi aur usko sahi jagah load karta hai. Isse unnecessary memory use nahi hoti.

Question: Process Management

Answer:

Process management ko kitchen example se samajh lo. CPU stove hai aur OS chef. Chef ko multiple dishes (programs) banani hoti hain. Wo dhyan rakhta hai ki koi dish zyada time na le aur har dish ko chance mile. Isi tarah OS sab programs ko proper scheduling deta hai taaki system smoothly chale aur sab processes efficiently execute ho sake.

Question: Security and Privacy

Answer:

Security: OS computer ko unauthorized users, viruses aur hackers se protect karta hai. Firewall, antivirus jaise features system ko safe rakhte hain.

Privacy: OS aapki personal information ko hide aur protect karta hai, bilkul ghar ke lock ki tarah jaha sirf aap hi access kar sakte ho.

Question: Resource Management

Answer:

OS system ke sare resources different processes me share karta hai. CPU time scheduling algorithms se manage hota hai, memory allocation OS handle karta hai, aur I/O devices ka control bhi OS ke paas hota hai. Ye ensure karta hai ki har resource sahi process ko sahi time par mile.

Question: User Interface

Answer:

OS user ko system use karne ka interface deta hai. Ye **CLI** (command line) ya **GUI** (mouse-based windows, menus) ho sakta hai. GUI users ke liye easy aur interactive hota hai.

Question: Networking

Answer:

OS network communication handle karta hai, jaise internet connect karna, data packets bhejna—receive

karna, aur network connections manage karna.

Question: Error Handling

Answer:

OS CPU, memory aur I/O devices me aane wale errors detect aur fix karta hai. Ye system ko deadlock se bachata hai, bugs track karta hai aur security breaches ko bhi prevent karta hai.

Question: Time Management

Answer:

Time management ko traffic light example se samajh sakte hain. Traffic light OS ki tarah kaam karti hai:

- **Red light = Simple queue** (program wait kar raha hai)
- **Yellow light = Ready queue** (program ready hai, bas chance ka wait)
- **Green light = Execution** (program run ho raha hai)

Jaise traffic light time-time par signal change karke sab gaadiyon ko smooth movement deti hai, waise hi OS time slices dekar sab programs ko fair chance deta hai taaki system smoothly chal sake.